WORKED SOLUTION

STEP I 2015, Question 1

Pure • Curve sketching

STEP I 2015, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s1-q1

The key idea

Factorise everything you can. The quadratic factorises, so the zeros are immediate, and the derivative is again e^x times a quadratic that factorises, so the turning points are immediate too. Once the sketch is right, counting solutions of "curve $= k$" is just sliding a horizontal line up and down and watching how many times it meets the curve. For part (ii), notice that the new function is the old one evaluated at x^2, so we can reuse almost everything.

Solution

Part (i)

Let $f(x) = e^x(2x^2 - 5x + 2) = e^x(2x - 1)(x - 2)$.

Zeros and sign. Since $e^x > 0$, $f(x) = 0$ exactly when $x = \frac{1}{2}$ or $x = 2$. Also f is positive for $x < \frac{1}{2}$ and $x > 2$, and negative on $(\frac{1}{2}, 2)$. The curve crosses the y-axis at $f(0) = 2$.

Turning points. By the product rule,

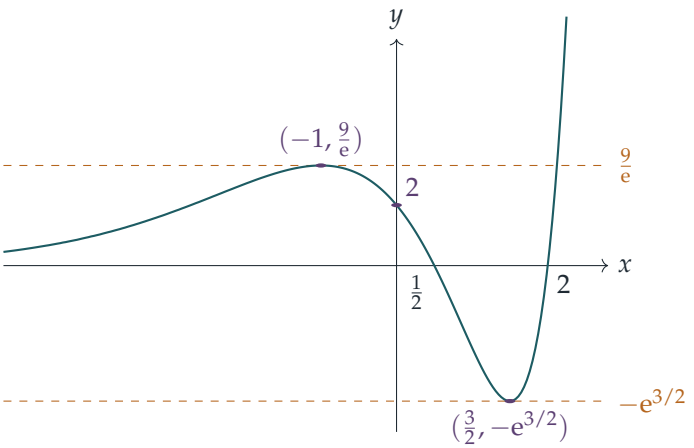
$$f'(x) = e^x(2x^2 - 5x + 2) + e^x(4x - 5) = e^x(2x^2 - x - 3) = e^x(2x - 3)(x + 1).$$

So $f'(x) = 0$ at $x = -1$ and $x = \frac{3}{2}$. The sign of f' is the sign of $(2x - 3)(x + 1)$: positive for $x < -1$, negative on $(-1, \frac{3}{2})$, positive for $x > \frac{3}{2}$. Hence f increases, then decreases, then increases, and

$$f(-1) = e^{-1}(2 + 5 + 2) = \frac{9}{e} \approx 3.31 \qquad \text{(local maximum)},$$

$$f(\tfrac{3}{2}) = e^{3/2}(\tfrac{9}{2} - \tfrac{15}{2} + 2) = -e^{3/2} \approx -4.48 \qquad \text{(local minimum)}.$$

Behaviour at the ends. As $x \to \infty$ both factors grow, so $f(x) \to \infty$. As $x \to -\infty$, the given fact (with $n = 2, 1, 0$) shows that each of $2x^2e^x$, $-5xe^x$ and $2e^x$ tends to 0, so $f(x) \to 0$. Since $f > 0$ for $x < \frac{1}{2}$, the curve approaches the negative x-axis from above: the x-axis is an asymptote.



Counting solutions of $f(x) = k$. The number of solutions is the number of times the horizontal line $y = k$ meets the curve. Going through the three monotonic pieces of the curve:

- on $(-\infty, -1]$, f increases strictly and takes every value in $(0, \frac{9}{e}]$ exactly once (it never reaches 0);
- on $[-1, \frac{3}{2}]$, f decreases strictly from $\frac{9}{e}$ to $-e^{3/2}$, taking each value in between exactly once;
- on $[\frac{3}{2}, \infty)$, f increases strictly from $-e^{3/2}$ to ∞, taking each value exactly once.

Adding up the contributions (and taking care not to count $x = -1$ or $x = \frac{3}{2}$ twice):

$k < -e^{3/2}$	no solutions
$k = -e^{3/2}$	one solution ($x = \frac{3}{2}$)
$-e^{3/2} < k \leq 0$	two solutions
$0 < k < \frac{9}{e}$	three solutions
$k = \frac{9}{e}$	two solutions ($x = -1$ and one with $x > 2$)
$k > \frac{9}{e}$	one solution

Ask yourself Why does $k = 0$ give two solutions, not three?

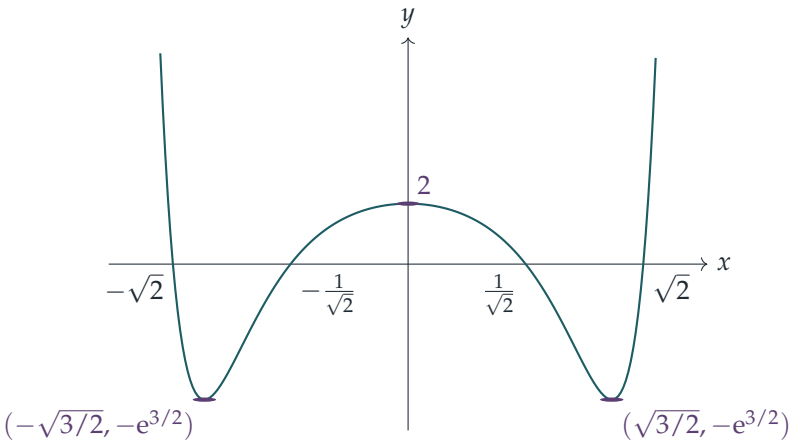
The x-axis is an asymptote on the left, and $f(x) > 0$ for every $x < \frac{1}{2}$, so the far-left branch never actually reaches 0. The only solutions of $f(x) = 0$ are $x = \frac{1}{2}$ and $x = 2$. For any small $k > 0$, though, the left branch does reach height k, which is why the count jumps to three.

Part (ii)

Let $g(x) = e^{x^2}(2x^4 - 5x^2 + 2)$. Then $g(x) = f(x^2)$, so g is even and its graph is symmetric in the y-axis. As x runs over $[0, \infty)$, $u = x^2$ runs over $[0, \infty)$ increasingly, so for $x \geq 0$ the graph of g just retraces the part of the graph of f with $u \geq 0$:

- $g(0) = f(0) = 2$;
- zeros where $x^2 = \frac{1}{2}$ or $x^2 = 2$, that is $x = \pm\frac{1}{\sqrt{2}}$ and $x = \pm\sqrt{2}$;
- by the chain rule, $g'(x) = 2x\,f'(x^2) = 2x\,\mathrm{e}^{x^2}(2x^2 - 3)(x^2 + 1)$, which vanishes only at $x = 0$ and $x = \pm\sqrt{3/2}$. (The turning point of f at $u = -1$ has disappeared, since $x^2 = -1$ has no real solution.)

For $0 < x < \sqrt{3/2}$ we have $g'(x) < 0$, and for $x > \sqrt{3/2}$ we have $g'(x) > 0$. So $(0, 2)$ is a local maximum (by symmetry $g' > 0$ just to the left of 0), and $x = \pm\sqrt{3/2}$ are minima with value $f(\frac{3}{2}) = -\mathrm{e}^{3/2}$. Finally $g(x) \to \infty$ as $x \to \pm\infty$, very rapidly because of the e^{x^2} factor. The curve is a "W" shape:



Checks

- $f(-1) = 9/\mathrm{e} \approx 3.311$ and $f(1.5) = -\mathrm{e}^{1.5} \approx -4.482$, and evaluating f at a few nearby points confirms that these are a maximum and a minimum.
- Near $x = 0$, $g(x) = f(x^2) \approx f(0) + f'(0)x^2 = 2 - 3x^2$, a downward parabola, which confirms the local maximum at $(0, 2)$.
- The root counts were confirmed by computer for sample values of k in each range (for example $k = 2$ gives three roots and $k = -1$ gives two).
- $f(-40) \approx 1.4 \times 10^{-14}$: positive, and tiny, as the asymptote predicts.

Common slips

- Counting three solutions when $k = 0$, by forgetting that the curve only approaches the x-axis on the left.
- Missing the boundary cases $k = \frac{9}{\mathrm{e}}$ and $k = -\mathrm{e}^{3/2}$, where the line is tangent to the curve and two solutions merge into one.

- In part (ii), keeping a turning point corresponding to $x^2 = -1$.

Beyond STEP

The counting argument rests on the intermediate value theorem together with strict monotonicity: a continuous strictly increasing function on an interval hits each value between its endpoint values exactly once. University analysis courses prove these facts carefully, along with the limit $x^n e^x \to 0$ that you were allowed to assume. Abbott's *Understanding Analysis* is a very readable place to see this done.

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